

1.



SIMATS
ENGINEERING

SIMATS Online Programming Lab

C++ PROGRAMMING

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QUESTIONS 19 / 19 completed

Concatenate ▾

CONCATENATE

Write a C++ program to overload a function to concatenate two strings and two character arrays separately.

PROGRAM EDITOR C++ ▾ Run Save

```
1 #include <iostream>
2 #include <string>
3 #include <cstring>
4 using namespace std;
5
6 // Overloaded function for std::string
7 string concatenate(const string& str1, const string& str2) {
8     return str1 + str2;
9 }
10
11 // Overloaded function for character arrays
12 string concatenate(const char str1[], const char str2[]) {
13     string temp = str1;
14     temp += str2;
15     return temp;
16 }
17
18 int main() {
19     string s1, s2;
20     char c1[100], c2[100];
21 }
```

INPUT

Hello World
SIMATS Programming

2.



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QUESTIONS 19 / 19 completed

Area

AREA

Write a C++ program to overload a function to calculate the area of a square, a rectangle, and a circle separately.

PROGRAM EDITOR

C++

Run Save

```

1 #include <iostream>
2 using namespace std;
3
4 // Overloaded function to calculate the area of a square
5 int area(int side) {
6     return side * side;
7 }
8
9 // Overloaded function to calculate the area of a rectangle
10 double area(double length, double breadth) {
11     return length * breadth;
12 }
13
14 // Overloaded function to calculate the area of a circle
15 double area(double radius) {
16     return 3.14159 * radius * radius;
17 }
18
19 int main() {
20     int side;
21

```

INPUT

5
 4.0 6.0
 3.0

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3.



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Sum

SUM

Write a C++ program to overload a function to calculate the sum of two matrices and two arrays separately.

PROGRAM EDITOR

C++

Run Save

```

1 #include <iostream>
2 using namespace std;
3
4 // Overloaded function to calculate the sum of two 1D arrays
5 void calculateSum(int a[], int b[], int size) {
6     cout << "Array Sum: ";
7     for (int i = 0; i < size; i++) {
8         cout << a[i] + b[i] << " ";
9     }
10    cout << endl;
11 }
12
13 // Overloaded function to calculate the sum of two 2D matrices
14 void calculateSum(int a[10][10], int b[10][10], int rows, int cols) {
15     cout << "Matrix Sum:" << endl;
16     for (int i = 0; i < rows; i++) {
17         for (int j = 0; j < cols; j++) {
18             cout << a[i][j] + b[i][j] << " ";
19         }
20         cout << endl;
21     }
22 }

```

INPUT

3
 1 2 3
 4 5 6
 2 2
 1 2
 3 4
 5 6
 7 8

4.

QUESTIONS 19 / 19 completed

Arrays

ARRAYS

Write a C++ program to overload a function to print the contents of an integer array, a double array, and a character array separately.

PROGRAM EDITOR

C++ Run Save

```

1 #include <iostream>
2 using namespace std;
3
4 // Overloaded function to print integer array
5 void printArray(int arr[], int size) {
6     cout << "Integer Array: ";
7     for (int i = 0; i < size; i++) {
8         cout << arr[i] << " ";
9     }
10    cout << endl;
11 }
12
13 // Overloaded function to print double array
14 void printArray(double arr[], int size) {
15     cout << "Double Array: ";
16     for (int i = 0; i < size; i++) {
17         cout << arr[i] << " ";
18     }
19     cout << endl;
20 }
21

```

INPUT

```

4
1 2 3 4
3
1.1 2.2 3.3
5
a b c d e

```

5.



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C++ PROGRAMMING

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QUESTIONS 19 / 19 completed

Factorial

FACTORIAL

Write a C++ program to overload a function to find the factorial of an integer number and the factorial of a floating-point number separately.

PROGRAM EDITOR

C++ Run Save

```

1 #include <iostream>
2 #include <cmath>
3 using namespace std;
4
5 // Function to calculate factorial for integers
6 long long factorial(int n) {
7     long long fact = 1;
8     for (int i = 1; i <= n; i++) {
9         fact *= i;
10    }
11    return fact;
12 }
13
14 // Overloaded function to calculate factorial (gamma function approximation) for float
15 double factorial(double n) {
16     return tgamma(n + 1);
17 }
18
19 int main() {
20     int i_num;
21

```

INPUT

```

5
4.5

```